

CHEN WEI LEE

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SUMMARY

Applied AI Engineer with production experience building LLM-powered systems. Deployed a public **Self-Evaluating Agentic RAG** and shipped a **multi-channel AI assistant with vector memory** at Inside Advisory. Fresh Graduate from University of Malaya (UM), 2 weeks' notice period.

EDUCATION

UNIVERSITY OF MALAYA

Bachelor of Computer Science (Artificial Intelligence)
CGPA: 3.80/4.00

Kuala Lumpur, Malaysia
Oct 2022 – March 2026

EXPERIENCE

INSIDE ADVISORY SDN BHD

Full-stack Developer & AI Engineer

Puchong, Malaysia
Aug 2025 – Present

A business advisory and software consultancy (10 – 20 employees) specializing in enterprise AI integration and custom SaaS development.

Core Responsibilities:

- Built Inside Assistant, a production multi-channel AI system (WhatsApp/Web/Lark) on Next.js with **identity-scoped vector memory** using **Supabase pgvector** and OpenAI text-embedding-3-small. Migrated LLM backend from self-hosted Claude Proxy to **BytePlus API** for cost efficiency. Serves **8-10 internal team users**.
- Prototyped **AI Conversational Lead Intelligence**, a real-time WhatsApp webhook pipeline with **multi-criteria LLM-based lead scoring**, **Groq Whisper** voice/image transcription for CRM intelligence.

Key Achievements & Impact:

- Delivered 3 client SaaS applications end-to-end on **Next.js** and **Supabase** generating RM 5k MRR across AV/IoT enterprise, steel manufacturing, and education verticals including **ERP with 10+ modules serving 20 daily users**, a FIFO manufacturing system with QR-driven cutting operations, and a coding school management platform. Additionally automated financial pipelines (P&L, balance sheet) using **n8n** and Node.js/Express with a live dashboard view.
- Maintain 3 pre-existing production SaaS systems (pest control, fire safety, concrete manufacturing) including bug fixes, feature requests, user support and performance triage across **TypeScript/Node.js** codebases.

DELL TECHNOLOGIES

Software Engineer Intern

Cyberjaya, Malaysia
Jul 2024 – Jan 2025

Global tech company providing enterprise hardware, IT infrastructure, and monitoring solutions.

- Independently developed key modules for an **agentless Database Monitoring System** to automate real-time infrastructure tracking and performance monitoring across **Oracle environments**.
- Expanded centralized monitoring architecture to natively support **3 distinct database types**, successfully eliminating the need for manual resource checks and complex client-side agent installations.
- Designed and implemented an integrated alerting mechanism featuring critical event categorization, significantly improving resource visibility and cross-team incident response times during infrastructure faults.

PROJECTS

Self-Evaluating Agentic RAG | [Live Demo](#)

- Deployed public Streamlit RAG chatbot with **BYOK (Bring Your Own Key)** and **built-in evaluation workflow**, users curate golden Q&A datasets and run on-demand evaluations tracking **Hit@k, MRR, keyword pass rate, and refusal accuracy**.
- Engineered hybrid retrieval: dense ChromaDB (sentence-transformers, MiniLM 384-dim) + BM25 lexical (rank-bm25), fused with **Reciprocal Rank Fusion (k=60)**, reranked with **BGE cross-encoder**. **Parent-child chunking** at 256/1024-token boundaries.
- Orchestrated agentic function-calling loop via **LangGraph StateGraph** with LLM, plus direct SDK peek-then-stream for per-token streaming UX.

Task-oriented Dialogue System (FYP)

- Multi-turn conversational AI in Python + Flask with **structured Dialogue State Tracker** and modular NLP for intent classification and slot filling; **hybrid NLU routing** prioritizing Rasa NLU for speed with LLM Fallback for complex or out-of-scope queries.

TECHNICAL SKILLS

AI/LLM: RAG · Vector Databases (ChromaDB, pgvector) · Embedding Models (OpenAI) · BGE Cross-Encoder Reranker · Hybrid Search (Dense + BM25) · Reciprocal Rank Fusion (RRF) · LangGraph · LangChain · Prompt Engineering · Function Calling · Agent Orchestration · RAG Evaluation (Hit@k, MRR) · Semantic Search

LLM APIs: BytePlus · Google Gemini · Anthropic Claude · OpenAI · Groq

Full-Stack: Python · TypeScript · JavaScript · SQL · Next.js · React · Node.js · Flask · Streamlit · Tailwind CSS · React Hook Form · PWA

Databases / Infra: PostgreSQL (Supabase) · pgvector · ChromaDB · Redis (Upstash) · Row-Level Security · Vercel · n8n · Git · GitHub · Zeabur

Testing / Integrations: Playwright · pytest · Vitest · WhatsApp Evolution API · Lark SDK · Google Cloud APIs · Groq Whisper · BillPlz

Languages: Fluent in Mandarin, English; Bahasa Melayu